# Darin Tsui

(909) 306-4398 | darint@gatech.edu | darintsui.github.io | darintsui | darintsui

## Experience

**Graduate Research Assistant** *Atlanta, GA*
AI ML and Information Research (AMIR) Group *Jun. 2023 - Present*
- Designed algorithms to achieve up to a 1000-fold reduction in the amortized computation of Shapley interactions.
- Applied Walsh-Hadamard transforms on variational autoencoders to extract explainable features.

**Medical Imaging Data Engineering Intern** *San Diego, CA*
Surgalign *Jan. 2023 - Aug. 2023*
- Designed an internal MRI image processing application using Python that reduced preprocessing time per sample.
- Assisted in developing ground truth dataset of imaging data to assess the effectiveness of deep learning models.
- Validated deep learning models against surgeon-generated data using Sørensen–Dice coefficient statistical testing.

**Development Engineer** *La Jolla, CA*
Integrated Systems Neuroengineering Laboratory *Jun. 2022 - Jul. 2023*
- Developed feature extraction and machine learning pipeline for bioelectronic COVID-19 detection using scikit-learn.
- Achieved accuracies of 98.5% when detecting COVID-19 proteins, improving classification by 30.1%.
- Published first-author 4-page paper to the Conference of the IEEE Engineering in Medicine and Biology Society (EMBC).

**Research Lead** *La Jolla, CA*
Talke Biomedical Device Laboratory *Dec. 2021 - Aug. 2023*
- Designed low-cost vision system for minimally invasive surgery using OpenCV with fiducial markers.
- Implemented Kalman filtering to achieve sub-millimeter error in design-validation testing.
- Published first-author papers to IEEE and the ASME Annual Conference on Information Storage and Processing Systems.

## Selected Publications

**Darin Tsui**, Aryan Musharaf, Justin S. Kang, Yigit Efe Erginbas, and Amirali Aghazadeh. SHAP zero explains biological sequence models with near-zero marginal cost for future queries. *arXiv preprint arXiv:2410.19236*, 2025.

**Darin Tsui** and Amirali Aghazadeh. On recovering higher-order interactions from protein language models. In *International Conference on Learning Representations (ICLR) 2024 Generative and Experimental Perspectives for Biomolecular Design (GEM) workshop*, 2024.

**Darin Tsui**, Kunal Talreja, and Amirali Aghazadeh. Efficient algorithm for sparse Fourier transform of generalized $q$-ary functions. *arXiv preprint arXiv:2501.12365*, 2025.

## Leadership

**President** *San Diego, CA*
IEEE at UC San Diego *May 2022 - May 2023*
- Managed operations for UC San Diego's 350+ student body by communicating with and delegating tasks to officers.
- Co-founded IEEE's Supercomputing Team with the San Diego Supercomputing Center (SDSC), increasing membership count by 10%.
- Hosted technical workshops on deep learning and classical machine learning by explaining mathematical concepts with relatable examples.

## Education

**Georgia Institute of Technology** *Atlanta, GA*
Ph.D. in Electrical and Computer Engineering *Aug. 2023 - Jun. 2028 (Expected)*

**University of California San Diego** *San Diego, CA*
Bachelor of Science in Bioengineering, GPA 3.919 *Sep. 2019 - Jun. 2023*

## Skills

**Programming** Python, MATLAB, Bash (Linux Shell Scripting)
**Libraries** PyTorch, TensorFlow, Scikit-learn, OpenCV, Scipy, Matplotlib, Numpy, Pandas
**Relevant Coursework** Statistical Learning, Neural Networks and Deep Learning, Bioinformatics Statistical Analysis